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Systems and methods for controlling a vehicle using high precision and high recall detection

Abstract

This disclosure provides systems and methods for detecting and tracking objects or obstacles in an environment of an autonomous vehicle. The method may include receiving data from a set of sensors, wherein the data represents objects or obstacles in an environment of the autonomous vehicle; and using a processor: generating high precision detection data based on the received data; identifying, from the high precision detection data, a set of objects that are classifiable by at least one known classifier; generating high recall detection data based on the received data; identifying from the high recall detection data a set of obstacles; and performing an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles.

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Background/Summary

FIELD

(1) This disclosure relates generally to systems and methods for detecting and tracking objects or obstacles in an environment of an autonomous vehicle based on high precision detection and high recall detection.

BACKGROUND

(2) Autonomous vehicles refer to vehicles that replace human drivers with sensors, computer-implemented intelligence, and other automation technology. Autonomous vehicles can be used to aid in the transport of passengers or items from one location to another. Such vehicles may operate in a fully autonomous mode where passengers may provide some initial input, such as a pickup or destination location, and the vehicle maneuvers itself to that location. While doing so, the safety of the passengers and the vehicle is an important consideration. Thus, there should be a high degree of confidence that autonomous vehicles will not collide with rare things or things the perception system has never seen before, while performing well in nominal cases with common and known object types.

(3) Therefore, there is a need for efficient vehicle motion planning systems and methods.

SUMMARY

(4) This disclosure addresses the above need in a number of aspects. In one aspect, this disclosure provides a method for detecting and tracking objects or obstacles in an environment of an autonomous vehicle, comprising: receiving data from a set of sensors, wherein the data represents objects or obstacles in an environment of the autonomous vehicle; and using a processor: (a) generating high precision detection data based on the received data; (b) identifying, from the high precision detection data, a set of objects that are classifiable by at least one known classifier; (c) generating high recall detection data based on the received data; (d) identifying from the high recall detection data a set of obstacles; and (e) performing an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles.

(5) In some embodiments, the operation comprises jointly optimizing the high precision detection data of the objects and the high recall detection data of the obstacles, when the autonomous vehicle is performing fallback maneuvers.

(6) In some embodiments, the operation comprises determining a cover value between an obstacle in the high recall detection data and an object in the high precision detection data, by dividing area of intersection of the obstacle and the object by area of the obstacle.

(7) In some embodiments, the operation comprises removing an obstacle from the high recall detection data if the cover value is greater than a threshold cover value. In some embodiments, the operation comprises removing an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data.

- (8) In some embodiments, if the cover value is between 1.0 and the threshold cover value, the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.
- (9) In some embodiments, the threshold cover value is 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more. In some embodiments, the threshold cover value is 0.5, 0.6, 0.7, 0.8, or 0.9.
- (10) In some embodiments, if a protrusion on the obstacle is the only difference between the obstacle and the object, the operation comprises associating the protrusion with the object and removing the obstacle from the high recall detection data.
- (11) In some embodiments, if the cover value is smaller than the threshold cover value, the operation comprises maintaining respective representations of the obstacle and the object.
- (12) In some embodiments, if the cover value is smaller than the threshold cover value and if the obstacle is associated with the object, the operation comprises integrating one or more characteristics of the object into characteristics of the obstacle while maintaining respective representations of the obstacle and the object.
- (13) In another aspect, this disclosure provides a method of controlling an autonomous vehicle. The method comprises: detecting and tracking objects or obstacles in an environment of the autonomous vehicle according to the method as described herein; determining a candidate trajectory for the autonomous vehicle based on the high precision detection data and/or the high recall detection data; and controlling the autonomous vehicle according to the candidate trajectory.
- (14) In yet another aspect, this disclosure provides a system for detecting and tracking objects or obstacles in an environment of an autonomous vehicle. In some embodiments, the system comprises a set of sensors, configured to receive data that represents objects or obstacles in an environment of the autonomous vehicle; and a processor, configured to: (i) generate high precision detection data based on the received data; (ii) identify, from the high precision detection data, a set of objects that are classifiable by at least one known classifier; (iii) generate high recall detection data based on the received data; (iv) identify from the high recall detection data a set of obstacles; and (v) perform an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles.
- (15) In some embodiments, the processor is configured to jointly optimize the high precision detection data of the objects and the high recall detection data of the obstacles, when the autonomous vehicle is performing fallback maneuvers.
- (16) In some embodiments, the processor is configured to determine a cover value between an obstacle in the high recall detection data and an object in the high precision detection data, by dividing area of intersection of the obstacle and the object by area of the obstacle.
- (17) In some embodiments, the processor is configured to remove an obstacle from the high recall detection data if the cover value is greater than a threshold cover value.
- (18) In some embodiments, the processor is configured to remove an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data.
- (19) In some embodiments, if the cover value is between 1.0 and the threshold cover value, the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.
- (20) In some embodiments, the threshold cover value is 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more. In some embodiments, the threshold cover value is 0.5, 0.6, 0.7, 0.8, or 0.9.
- (21) In some embodiments, if a protrusion on the obstacle is the only difference between the obstacle and the object, the processor is further configured to associate the protrusion with the object and removing the obstacle from the high recall detection data.
- (22) In some embodiments, if the cover value is smaller than the threshold cover value, the processor is further configured to maintain respective representations of the obstacle and the object.

(23) In some embodiments, if the cover value is smaller than the threshold cover value and if the obstacle is associated with the object, the processor is further configured to integrate one or more characteristics of the object into characteristics of the obstacle while maintaining respective representations of the obstacle and the object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing high precision detection and high recall detection for detecting objects of known classes and unknown objects by an autonomous vehicle, according to various embodiments of the present disclosure.
- (2) FIG. 2 illustrates an example system for controlling a vehicle based on a combination of high precision detection and high recall detection, according to various embodiments of the present disclosure.
- (3) FIG. 3a is a flowchart of an example method for controlling a vehicle based on attributes determined by data received from an individual sensor, according to various embodiments of the present disclosure.
- (4) FIG. 3b is a flowchart of an example method for controlling a vehicle based on fusion of sensor data received from a set of sensors, according to various embodiments of the present disclosure.
- (5) FIG. 3c is a flowchart of an example method for controlling a vehicle based on a combination of high precision detection and high recall detection, according to various embodiments of the present disclosure.
- (6) FIG. 3d is a flowchart of an example method for detecting and tracking objects or obstacles in an environment of an autonomous vehicle, according to various embodiments of the present disclosure.
- (7) FIG. 4 illustrates example elements of a computing device, according to various embodiments of the present disclosure.
- (8) FIG. 5 illustrates an example architecture of a vehicle, according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

- (9) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. These terms are merely intended to distinguish one component from another component, and the terms do not limit the nature, sequence or order of the constituent components.
- (10) It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Throughout the specification, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.
- (11) In addition, the terms “unit,” “-er,” “-or,” and “module” described in the specification mean units for processing at least one function and operation, and can be implemented by hardware components or software components and combinations thereof.
- (12) In this document, when terms such as “first” and “second” are used to modify a noun, such use is simply intended to distinguish one item from another, and is not intended to require a sequential order unless specifically stated.

(13) In addition, terms of relative position such as “vertical” and “horizontal,” or “front” and “rear,” when used, are intended to be relative to each other and need not be absolute, and only refer to one possible position of the device associated with those terms depending on the device's orientation.

(14) An “electronic device” or a “computing device” refers to a device that includes a processor and memory. Each device may have its own processor and/or memory, or the processor and/or memory may be shared with other devices as in a virtual machine or container arrangement. The memory will contain or receive programming instructions that, when executed by the processor, cause the electronic device to perform one or more operations according to the programming instructions.

(15) The terms “memory,” “memory device,” “computer-readable storage medium,” “data store,” “data storage facility,” and the like each refer to a non-transitory device on which computer-readable data, programming instructions or both are stored. Except where specifically stated otherwise, the terms “memory,” “memory device,” “computer-readable storage medium,” “data store,” “data storage facility,” and the like are intended to include single device embodiments, embodiments in which multiple memory devices together or collectively store a set of data or instructions, as well as individual sectors within such devices.

(16) The terms “processor” and “processing device” refer to a hardware component of an electronic device that is configured to execute programming instructions. Except where specifically stated otherwise, the singular term “processor” or “processing device” is intended to include both single-processing device embodiments and embodiments in which multiple processing devices together or collectively perform a process.

(17) The terms “instructions” and “programs” may be used interchangeably herein. The instructions may be stored in object code format for direct processing by the processor, or in any other computing device language, including scripts or collections of independent source code modules that are interpreted on demand or compiled in advance. Functions, methods, and routines of the instructions are explained in more detail below. The instructions may be any set of instructions to be executed directly (such as machine code) or indirectly (such as scripts) by the processor. For example, the instructions may be stored as computing device code on the computing device-readable medium.

(18) The term “data” may be retrieved, stored or modified by processors in accordance with a set of instructions. For instance, although the claimed subject matter is not limited by any particular data structure, the data may be stored in computing device registers, in a relational database as a table having a plurality of different fields and records, XML documents or flat files. The data may also be formatted in any computing device-readable format.

(19) The term “module” or “unit” refers to a set of computer-readable programming instructions, as executed by a processor, that cause the processor to perform a specified function.

(20) The term “vehicle,” or other similar terms, refers to any motor vehicles, powered by any suitable power source, capable of transporting one or more passengers and/or cargo. The term “vehicle” includes, but is not limited to, autonomous vehicles (i.e., vehicles not requiring a human operator and/or requiring limited operation by a human operator), automobiles (e.g., cars, trucks, sports utility vehicles, vans, buses, commercial vehicles, etc.), boats, drones, trains, and the like.

(21) The term “autonomous vehicle,” “automated vehicle,” “AV,” or “driverless vehicle,” as used herein, refers to a vehicle capable of implementing at least one navigational change without driver input. A “navigational change” refers to a change in one or more of steering, braking, or acceleration of the vehicle. To be autonomous, a vehicle need not be fully automatic (e.g., fully operation without a driver or without driver input). Rather, an autonomous vehicle includes those that can operate under driver control during certain time periods and without driver control during other time periods. Autonomous vehicles may also include vehicles that control only some aspects of vehicle navigation, such as steering (e.g., to maintain a vehicle course between vehicle lane

constraints), but may leave other aspects to the driver (e.g., braking). In some cases, autonomous vehicles may handle some or all aspects of braking, speed control, and/or steering of the vehicle. Autonomous vehicles may be any type of vehicle including, but not limited to, cars, trucks, motorcycles, busses, recreational vehicles, agricultural vehicles, construction vehicles etc. According to various embodiments, autonomous vehicles may include a throttle control system and a braking system. Autonomous vehicles may include one or more engines and/or one or more computing devices. The one or more computing devices may be separate from the automated speed control system or the braking system. Additionally, the computing device may include a processor and/or a memory. The memory may be configured to store programming instructions that, when executed by the processor, are configured to cause the processor to perform one or more tasks. In certain embodiments, autonomous vehicles may include a receiver configured process the communication between autonomous vehicles and a teleoperation system.

(22) The term “trajectory” or “map” is used broadly to include, for example, a motion plan or any path or route from one place to another; for instance, a path from a pickup location to a drop off location.

(23) Although exemplary embodiment is described as using a plurality of units to perform the exemplary process, it is understood that the exemplary processes may also be performed by one or plurality of modules. Additionally, it is understood that the term controller/control unit refers to a hardware device that includes a memory and a processor and is specifically programmed to execute the processes described herein. The memory is configured to store the modules, and the processor is specifically configured to execute said modules to perform one or more processes which are described further below.

(24) Further, the control logic of the present disclosure may be embodied as non-transitory computer-readable media on a computer-readable medium containing executable programming instructions executed by a processor, controller, or the like. Examples of computer-readable media include, but are not limited to, ROM, RAM, compact disc (CD)-ROMs, magnetic tapes, floppy disks, flash drives, smart cards, and optical data storage devices. The computer-readable medium can also be distributed in network-coupled computer systems so that the computer-readable media may be stored and executed in a distributed fashion such as by a telematics server or a Controller Area Network (CAN).

(25) Unless specifically stated or obvious from context, as used herein, the term “about” is understood as within a range of normal tolerance in the art, for example, within two standard deviations of the mean. About can be understood as within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.1%, 0.05%, or 0.01% of the stated value.

(26) Hereinafter, systems and methods for controlling a vehicle, according to embodiments of the present disclosure, will be described with reference to the accompanying drawings. In the drawings, the same reference numerals will be used throughout to designate the same or equivalent elements. In addition, a detailed description of well-known features or functions will be ruled out in order not to unnecessarily obscure the gist of the present disclosure.

(27) With reference to FIG. 1, autonomous vehicles, e.g., an autonomous vehicle **100**, are required to bring goods or passengers to desired locations safely. There must be a high degree of confidence that autonomous vehicles will not collide with rare things or things a perception system has never seen before, while performing well in nominal cases with common and known object types. However, analyzing the same objects or obstacles observed in both high precision detection and high recall detection can be time-consuming and require a significant amount of processing power. Thus, it is important for the autonomous vehicle **100** to efficiently generate trajectories by reducing duplicate detection or duplicate calculation of objects or obstacles of common and known object types **110** and objects or obstacles without class identification **120** (e.g., rare or unknown objects). To this end, the autonomous vehicle **100** may employ separate detectors for high precision detection and high recall detection, remove duplicate detections, and combine results in a motion

planning component.

(28) Referring now to FIG. 2, a system **200** may be implemented to control the autonomous vehicle **100** based on high precision detection and high recall detection of objects and obstacles in the environment of the autonomous vehicle **100**. The autonomous vehicle **100** may be any type of vehicle including, but not limited to, cars, trucks, motorcycles, buses, recreational vehicles, etc. The system can detect objects or obstacles in the environment of the autonomous vehicle **100** using both high precision detection and high recall detection, and the high precision detection and high recall detection are efficient enough to be executed in “real time” for the purposes of automatic driving. For instance, as described below, the system **200** may use high recall detection (higher sensitivity and lower accuracy) to identify any collidable objects or obstacles and generate initial object candidates, including objects or obstacles of common and known object types **110** and objects or obstacles without class identification **120** (e.g., rare or unknown objects). The system **200** may use high precision detection (higher accuracy), e.g., by using more accurate classifiers. In high recall detection **220**, the system **200** processes sensor data quickly with lower accuracy, while in high precision detection **210**, the system **200** processes sensor data at a slower speed but with higher accuracy. In addition, the system **200** can be implemented to process data from different kinds of sensors, e.g., LADAR, and video cameras, and combine the data from different sensors to improve overall object detection performance.

(29) The system **200** may perform high precision detection **210** to detect objects and obstacles of common and known classes **110**. The high precision detection **210** may be carried out by one or more image detectors (e.g., camera) and/or one or more point cloud detectors (e.g., LIDAR) tuned for high precision detection. The objects and obstacles of common and known classes **110** refer to objects or obstacles that can be classified by at least one known classifier (e.g., vehicle classifiers). For example, the objects and obstacles of common and known classes **110** can belong to any classification category, such as other vehicles, bicyclists, or pedestrians. In addition, the high precision detection may further identify contextual information about each object, for example, the speed and pose of the object, direction of movement, presence of other dynamic objects, and other information.

(30) After the objects and obstacles of common and known classes **110** are detected, the system **200** may perform object tracking **212** to track movement of the detected objects over time and maintain their identity (e.g., vehicles, bicyclists, pedestrians) to identify tracked high precision objects **214**. The tracked high precision objects **214** may include the closest vehicle in the same lane or different lanes as the autonomous vehicle **100**.

(31) The system **200** may additionally perform high recall detection **220** to detect objects and obstacles of common and known classes **110**. The high recall detection **220** may be carried out by point cloud clustering by LIDAR, stereo depth vision by RADAR, and/or monocular depth vision using learned low-level features by RADAR, tuned for high recall detection.

(32) The system **200** may further perform coverage filtering **222** to remove objects identified by the high recall detection **220** that match the tracked high precision objects **214**, resulting in filtered high recall objections **224**. Based on the information about the tracked high precision objects **214** and the filtered high recall objects **224**, the system **200** may perform object motion prediction and ego motion planning **230** and ego vehicle control **240** of the autonomous vehicle **100**.

(33) Referring to FIG. 3a, an example flowchart of a method **300** for controlling the autonomous vehicle is provided, in accordance with various embodiments of the present disclosure.

(34) At **301a**, the method **300** may include receiving data from a set of sensors, wherein the data represents objects or obstacles in the environment of the autonomous vehicle. In some embodiments, the set of sensors may include a two-dimensional object detector, a three-dimensional object detector, and/or an obstacle detector. In some embodiments, the set of sensors may include RADAR, LIDAR, camera, sonar, laser, or ultrasound.

(35) At **302a**, the method **300** may include identifying objects or obstacles from the received data.

For example, the method **300** may include identifying various objects or obstacles in the physical environment of the autonomous vehicle by an object detection system. The objects may belong to one or more classes, such as cars, trucks, buses, etc. In some embodiments, the object detection system may utilize a neural network, e.g., a deep convolutional neural network (DCNN), for object detection. In addition, the same neural network (e.g., deep convolutional neural network) can also predicts the centroid locations or other portions for each detected instance.

(36) At **303a**, the method **300** may include determining multiple sets of attributes of the objects or obstacles. In some embodiments, each set of attributes of the objects or obstacles may be determined based on data received by an individual sensor. In some embodiments, the individual sensor may be a RADAR, LIDAR, camera, sonar, laser, or ultrasound sensor. Different types of sensors may be used to determine different attributes of the objects or obstacles. For example, RADAR may be used to create 3D models and maps of objects and environments, and a camera may be used to determine attributes such as object classification information of the objects or obstacles. In some embodiments, the attributes of the objects or obstacles may include kinematic information, geometric information, or object classification information. In some embodiments, kinematic information may include position information, velocity information, and/or orientation information. In some embodiments, geometric information may include bounding boxes and/or object contours. In some embodiments, object classification information may include an object type.

(37) At **304a**, the method **300** may include determining a candidate trajectory for the autonomous vehicle based on the multiple sets of attributes of the objects or obstacles. At **305a**, the method **300** may include controlling the autonomous vehicle according to the candidate trajectory. A candidate trajectory should allow the autonomous vehicle to avoid objects or obstacles in the environment of the autonomous vehicle. However, different attributes of the objects or obstacles can lead to different strategies for tracking. For example, the autonomous vehicle may identify an object or obstacle based on point cloud data obtained from a LIDAR. The autonomous vehicle may determine that the object or obstacle does not require further tracking because the velocity of the object or obstacle meets the characteristics of an obstacle (e.g., a tree, a building, a traffic sign) instead of an object (e.g., a person, a vehicle).

(38) Referring now to FIG. **3b**, an example flowchart of a method **300** for controlling the autonomous vehicle based on high precision and high recall detection is provided, in accordance with various embodiments of the present disclosure. At **301b**, the method **300** may include receiving data from a set of sensors, wherein the data represents objects or obstacles in the environment of the autonomous vehicle. In some embodiments, the set of sensors may include a two-dimensional object detector, a three-dimensional object detector, and/or an obstacle detector. In some embodiments, the set of sensors may include RADAR, LIDAR, camera, sonar, laser, or ultrasound.

(39) At **302b**, the method **300** may include determining attributes of each of the objects or obstacles based on the received data from the set of sensors. As described above, the attributes of the objects or obstacles may include kinematic information, geometric information, or object classification information. In some embodiments, kinematic information may include position information, velocity information, and/or orientation information. In some embodiments, geometric information may include bounding boxes and/or object contours. In some embodiments, object classification information may include an object type.

(40) At **303b**, the method **300** may include integrating the attributes of each of the objects or obstacles. For example, the method **300** may process and synthesize all the information received in a process including sensor fusion of all sensor information received. In some embodiments, the method **300** may include performing sensor fusion for the received data from the set of sensors, such as two-dimensional object detectors, three-dimensional object detectors, and/or obstacle detectors. In some embodiments, the method **300** may include performing sensor fusion for the

received data from a set of object detectors. At **304b**, the integrated sensor data may be used for identification of objects or obstacles and/or determining one or more attributes of the objects or obstacles. The terms “integrate,” “fuse,” and “synthesize” are used interchangeably herein. Unlike the method illustrated in FIG. **3a**, the method as described in FIG. **3b** may integrate data received from two or more sensors, such that the data can be used to detect and/or track objects or obstacles. (41) In some embodiments, the method may include fusing received data from all sensors, such as vision sensors, LIDAR, and RADAR. In some embodiments, the sensors may include two stereo vision sensors, RADAR, and/or LIDAR. Stereo vision may use imaging technology to output bounded boxes of objects (e.g., cars, bikes, people, lanes, signs, and objects) found in the field of view of the cameras. RADAR systems may track locations and ranges of any objects that reflect RADAR—but cannot classify or identify those objects. The LIDAR system can recognize cars, bikes, or other objects and provides object classifications of the same.

(42) In some embodiments, the method may include analyze the fused data, including data from vision sensors, LIDAR, and/or RADAR, to determine a complete picture of objects surrounding the self-driving car.

(43) At **304b**, the method **300** may include determining a candidate trajectory for the autonomous vehicle based on the multiple sets of attributes of the objects or obstacles. At **305b**, the method **300** may include navigating the autonomous vehicle according to the candidate trajectory at least to avoid objects or obstacles in the environment of the autonomous vehicle.

(44) Referring now to FIG. **3c**, an example flowchart of a method **300** for controlling the autonomous vehicle **100** based on high precision and high recall detection is provided, in accordance with various embodiments of the present disclosure. At **311**, the method **300** may include receiving data from a set of sensors, wherein the data represents objects or obstacles in the environment of the autonomous vehicle. In some embodiments, the set of sensors may include a two-dimensional object detector, a three-dimensional object detector, and/or an obstacle detector. In some embodiments, the set of sensors may include RADAR, LIDAR, camera, sonar, laser, or ultrasound.

(45) At **312**, the method **300** may include generating high precision detection data based on the received data. In some embodiments, the method **300** may include generating the high precision detection data from the data received from one or more image detectors (e.g., camera) and/or one or more point cloud detectors (e.g., LIDAR) tuned for high precision detection. The high precision detection data may be generated from different kinds of sensors, e.g., LADAR, and video cameras, and combine the data from different sensors to improve overall object detection performance.

(46) At **313**, the method **300** may include identifying from the high precision detection data a first set of objects or obstacles that are classifiable by at least one known classifier (e.g., a vehicle classifier, a bicyclist classifier, a pedestrian classifier). For example, the objects and obstacles identified by a known classifier can belong to any classification category, such as other vehicles, bicyclists, or pedestrians. In addition, the method **300** may further include identifying contextual information about each object, for example, the speed and pose of the object, direction of movement, presence of other dynamic objects, and other information.

(47) At **314**, the method **300** may include tracking movement of one or more objects in the first set of objects or obstacles over time and maintaining identity (e.g., vehicles, bicyclists, pedestrians) of the tracked one or more objects in the first set of objects or obstacles. In some embodiments, the method **300** may include tracking movement of the one or more objects in the first set of objects or obstacles over a period of time when the one or more objects are detectable by the set of sensors. In some embodiments, the period is from about 100 ms to about 500 ms. In some embodiments, the period is about 300 ms.

(48) At **315**, the method **300** may include generating high recall detection data based on the received data. At **316**, the method **300** may include identifying from the high recall detection data a second set of objects or obstacles without using any classifier, e.g., a vehicle classifier, a bicyclist

classifier, or a pedestrian classifier). In some embodiments, the method **300** may include generating the high recall detection data from point cloud clustering by LIDAR, stereo depth vision by RADAR, and/or monocular depth vision using learned low-level features by RADAR. In some embodiments, the learned low-level features may include distance.

(49) In some embodiments, the method **300** may include generating the high precision detection data and the high recall detection data based on different subsets of data received from different sets of sensors. the method **300** may include performing sensor fusion for the different subsets of data received from different sets of sensors in order to generate the high detection data or the high recall detection data.

(50) At **317**, the method **300** may include filtering out objects, from the second set of objects or obstacles, that correspond to the tracked one or more objects in the first set of objects or obstacles, to obtain a filtered set of objects or obstacles. In some embodiments, the method **300** may include filtering out a set of points corresponding to the tracked one or more objects in the first set of objects or obstacles from at least one point cloud of the second set of objects or obstacles.

(51) At **318**, the method **300** may include determining a candidate trajectory for the autonomous vehicle to avoid at least the tracked one or more objects in the first set of objects or obstacles and the filtered set of objects or obstacles. At **319**, the method **300** may further include controlling the autonomous vehicle according to the candidate trajectory.

(52) Referring now to FIG. **3d**, an example flowchart of a method **300** for detecting and tracking objects or obstacles in an environment of an autonomous vehicle based on high precision and high recall detection is provided, in accordance with various embodiments of the present disclosure.

(53) At **321**, the method **300** may include receiving data from a set of sensors, wherein the data represents objects or obstacles in the environment of the autonomous vehicle. In some embodiments, the set of sensors may include a two-dimensional object detector, a three-dimensional object detector, and/or an obstacle detector. In some embodiments, the set of sensors may include RADAR, LIDAR, camera, sonar, laser, or ultrasound.

(54) At **322**, the method **300** may include generating high precision detection data based on the received data. Similarly, at **324**, the method **300** may include generating high recall detection data based on the received data.

(55) At **323**, the method **300** may include identifying, from the high precision detection data, a set of objects that are classifiable by at least one known classifier. At **325**, the method **300** may include identifying from the high recall detection data a set of obstacles. For example, the method **300** may include identifying various objects or obstacles in the physical environment of the autonomous vehicle by an object detection system. The objects may belong to one or more classes, such as cars, trucks, buses, etc. In some embodiments, the object detection system may utilize a neural network, e.g., a convolutional neural network (CNN), a deep convolutional neural network (DCNN), for object detection. In addition, the same neural network (e.g., deep convolutional neural network) can also predicts the centroid locations or other portions for each detected instance.

(56) At **326**, the method **300** may include performing an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles. In some embodiments, the operation may include merging/integrating the high precision detection data of an object with the high recall detection data of an obstacle.

(57) In some embodiments, the operation may include jointly optimizing the high precision detection data of the objects and the high recall detection data of the obstacles. For example, if the truck is in fallback, the pathway may be fallback to a shoulder or in-lane of a road. The operation may include keeping obstacles and objects as duplicate states (e.g., maintaining respective representations of obstacles and objections) and optimizing based on the state of the autonomous vehicle. The detections may also be jointly optimized relative to other detections. Each detection may have its own process to generate the characteristics or attributes (e.g., shape, speed, direction)

of the object or obstacle. The detections may share with other detections the characteristics or attributes that the detection has generated to add additional information to the other detections. The other detections, then, may use the additional information from the other detections to add to its prediction or confidence in predicting the characteristics of the object or obstacle that the detection is detecting. In this embodiment, each detection may have additional inputs from other detections in order for the detection to optimize the attributes or characteristics of the object or obstacle it is detecting.

(58) In some embodiments, the operation may include filtering out an obstacle from the high recall detection data if the obstacle is identical to or shares a high level of similarity with an object identified in the high precision detection data. To determine the level of similarity between an obstacle in the high recall detection data and an object in the high precision detection data, the method may include determining a cover value between an obstacle in the high recall detection data and an object in the high precision detection data, by dividing area of intersection of the obstacle and the object by area of the obstacle. For example, the cover value may be computed by dividing the area of intersection of the obstacle and the object in 2D overhead space (or 3D space) by the obstacle area. If the cover value is 1.0, the object is considered identical to the obstacle (or referred to as “the object is fully covered”).

(59) In some embodiments, the operation may include removing an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data. In some embodiments, if the cover value is smaller than the threshold cover value, the operation may include maintaining respective representations of the obstacle and the object.

(60) In some embodiments, the operation may include removing an obstacle from the high recall detection data if the cover value is greater than a threshold cover value. In some embodiments, the threshold cover value is 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more. In some embodiments, the threshold cover value is 0.5, 0.6, 0.7, 0.8, or 0.9.

(61) If the cover value is not 1.0, but the cover value is higher than a threshold cover value (e.g., 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more), then the obstacle and object are still treated as the same objects, but the object representation needs to be modified to fully explain or cover the obstacle. For example, the method may include dilating the bounding box of the object to cover the obstacle and deleting the obstacle. In some embodiments, if the cover value is between 1.0 and the threshold cover value (e.g., 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more), the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.

(62) In some embodiments, if a protrusion (e.g., something sticking off the obstacle) on the obstacle is the only difference between the obstacle and the object, the operation may include associating the protrusion with the object and removing the obstacle from the high recall detection data.

(63) In some embodiments, if the cover value is smaller than the threshold cover value (e.g., 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, or more) and if the obstacle is associated with the object, the operation may include integrating one or more characteristics of the object into characteristics of the obstacle while maintaining respective representations of the obstacle and the object. As used herein, the phrase “the obstacle is associated with the object” means that identification of the obstacle and the object comes from the same sensor measurements, and/or have reasonably high Intersection over Union (IOU) at various times, etc. For example, if an obstacle (e.g., a large piece of equipment having off the back of a truck) is determined as moving with an object (e.g., a truck), the obstacle will be treated as being “associated with” the object. Accordingly, some of characteristics (e.g., velocity, speed, distance, etc.) of the object will be imputed to the obstacle that is associated with the object.

(64) The method as disclosed above may be performed as part of a process for controlling an autonomous vehicle. In some embodiments, the method controlling an autonomous vehicle may

include detecting and tracking objects or obstacles in an environment of the autonomous vehicle according to the method as described herein. In some embodiments, the method may also include determining a candidate trajectory for the autonomous vehicle based on the high precision detection data and/or the high recall detection data. In some embodiments, the method may further include controlling the autonomous vehicle according to the candidate trajectory.

(65) In some embodiments, the operation may include mixing/merging characteristics or attributes of objects or obstacles from different detections based on worst case of what a planner wants protect against from collision. In one embodiment, the mixer may take individual characteristics or attributes from different detectors to create an instance of the object or obstacle that may incorporate the worst case attributes detected from the different detectors. Thus, the mixing function may create a new detection that is a combination of the different detectors that is the worst case attribute for a specific object or obstacle based upon how the planner would interact with the object or obstacle. For example, in the event that the camera detector, the LIDAR detector, and the Radar detector have all generated a detection of a specific object and each detector detects a different speed for the object, combining those different objects into a single object may then take the worst case scenario of the different speed detections. When that object is in the path, then the highest speed from the detection will be used. This is because the highest speed yields the minimum time to interaction with the object. If the object is in an adjacent lane, then the detected speed that is closest to the speed of the autonomous vehicle is the worst case scenario because that leaves the object closest for interaction with the autonomous vehicle for the most amount of time.

(66) In another embodiment, instead of the worst case scenario, the mixing function may take the value for the attribute that is most likely from the detectors, or predicted with the higher confidence between different detectors.

(67) Referring now to FIG. 4, an illustration of an example architecture for a computing device **400** is provided. The main computing system **210** or the secondary controlling system **220** of FIG. 1 may be the same as or similar to computing device **400**. As such, the discussion of computing device **400** is sufficient for understanding the main computing system **210** or the secondary controlling system **220** of FIG. 1, for example.

(68) Computing device **400** may include more or fewer components than those shown in FIG. 1. The hardware architecture of FIG. 4 represents one example implementation of a representative computing device configured to one or more methods and means for controlling the autonomous vehicle **100** in response to an abnormal condition of the autonomous vehicle **100**, as described herein. As such, the computing device **400** of FIG. 4 implements at least a portion of the method(s) described herein (for example, method **300** of FIGS. 3a-c).

(69) Some or all components of the computing device **400** can be implemented as hardware, software and/or a combination of hardware and software. The hardware includes, but is not limited to, one or more electronic circuits. The electronic circuits can include, but are not limited to, passive components (e.g., resistors and capacitors) and/or active components (e.g., amplifiers and/or microprocessors). The passive and/or active components can be adapted to, arranged to and/or programmed to perform one or more of the methodologies, procedures, or functions described herein.

(70) As shown in FIG. 4, the computing device **400** may include a user interface **402**, a Central Processing Unit (“CPU”) **406**, a system bus **410**, a memory **412** connected to and accessible by other portions of computing device **400** through system bus **410**, and hardware entities **414** connected to system bus **410**. The user interface can include input devices and output devices, which facilitate user-software interactions for controlling operations of the computing device **400**. The input devices may include, but are not limited to, a physical and/or touch keyboard **450**. The input devices can be connected to the computing device **400** via a wired or wireless connection (e.g., a Bluetooth® connection). The output devices may include, but are not limited to, a speaker **452**, a display **454**, and/or light-emitting diodes **456**.

(71) At least some of the hardware entities **414** perform actions involving access to and use of memory **412**, which can be a Random Access Memory (RAM), a disk drive and/or a Compact Disc Read Only Memory (CD-ROM), among other suitable memory types. Hardware entities **414** can include a disk drive unit **416** comprising a computer-readable storage medium **418** on which is stored one or more sets of instructions **420** (e.g., programming instructions such as, but not limited to, software code) configured to implement one or more of the methodologies, procedures, or functions described herein. The instructions **420** can also reside, completely or at least partially, within the memory **412** and/or within the CPU **406** during execution thereof by the computing device **400**. The memory **412** and the CPU **406** also can constitute machine-readable media. The term “machine-readable media,” as used here, refers to a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions **420**. The term “machine-readable media,” as used here, also refers to any medium that is capable of storing, encoding or carrying a set of instructions **420** for execution by the computing device **400** and that cause the computing device **400** to perform any one or more of the methodologies of the present disclosure.

(72) Referring now to FIG. 5, an example vehicle system architecture **500** for a vehicle is provided, in accordance with various embodiments of the present disclosure.

(73) The autonomous vehicle **100** of FIG. 1 can have the same or similar system architecture as shown in FIG. 5. Thus, the following discussion of vehicle system architecture **500** is sufficient for understanding the autonomous vehicle **100** of FIG. 1.

(74) As shown in FIG. 5, the vehicle system architecture **500** includes an engine, motor or propulsive device (e.g., a thruster) **502** and various sensors **504-518** for measuring various parameters of the vehicle system architecture **500**. In gas-powered or hybrid vehicles having a fuel-powered engine, the sensors **504-518** may include, for example, an engine temperature sensor **504**, a battery voltage sensor **506**, an engine Rotations Per Minute (RPM) sensor **508**, and/or a throttle position sensor **510**. If the vehicle is an electric or hybrid vehicle, then the vehicle may have an electric motor, and accordingly will have sensors such as a battery monitoring system **512** (to measure current, voltage and/or temperature of the battery), motor current **514** and voltage **516** sensors, and motor position sensors such as resolvers and encoders **518**.

(75) Operational parameter sensors that are common to both types of vehicles include, for example, a position sensor **534**, such as an accelerometer, gyroscope and/or inertial measurement unit; a speed sensor **536**; and/or an odometer sensor **538**. The vehicle system architecture **500** also may have a clock **542** that the system uses to determine vehicle time during operation. The clock **542** may be encoded into the vehicle onboard computing device **520**. It may be a separate device, or multiple clocks may be available.

(76) The vehicle system architecture **500** also may include various sensors that operate to gather information about the environment in which the vehicle is traveling. These sensors may include, for example, a location sensor **544** (for example, a Global Positioning System (GPS) device); object detection sensors such as one or more cameras **546**; a LIDAR sensor system **548**; and/or a radar and/or a sonar system **550**. The sensors also may include environmental sensors **552**, such as a precipitation sensor and/or ambient temperature sensor. The object detection sensors may enable the vehicle system architecture **500** to detect objects that are within a given distance range of the vehicle **500** in any direction, while the environmental sensors **552** collect data about environmental conditions within the vehicle's area of travel.

(77) During operations, information is communicated from the sensors to an onboard computing device **520**. The onboard computing device **520** may be configured to analyze the data captured by the sensors and/or data received from data providers, and may be configured to optionally control operations of the vehicle system architecture **500** based on the results of the analysis. For example, the onboard computing device **520** may be configured to control: braking via a brake controller **522**; direction via a steering controller **524**; speed and acceleration via a throttle controller **526** (in a

gas-powered vehicle) or a motor speed controller **528** (such as a current level controller in an electric vehicle); a differential gear controller **530** (in vehicles with transmissions); and/or other controllers.

(78) Geographic location information may be communicated from the location sensor **544** to the onboard computing device **520**, which may then access a map of the environment that corresponds to the location information to determine known fixed features of the environment such as streets, buildings, stop signs and/or stop/go signals. Captured images from the cameras **546** and/or object detection information captured from sensors such as LIDAR **548** are communicated from those sensors to the onboard computing device **520**. The object detection information and/or captured images are processed by the onboard computing device **520** to detect objects in proximity to the vehicle. Any known or to be known technique for making an object detection based on sensor data and/or captured images may be used in the embodiments disclosed in this document.

(79) The present disclosure is not to be limited in scope by the specific embodiments described herein. Indeed, various modifications of the invention in addition to those described herein will become apparent to those skilled in the art from the foregoing description and the accompanying figures. Such modifications are intended to fall within the scope of the appended claims.

Claims

1. A method for detecting and tracking objects or obstacles in an environment of an autonomous vehicle, comprising: receiving data from a set of sensors, wherein the data represents objects or obstacles in an environment of the autonomous vehicle; and using a processor to perform the steps of: generating high precision detection data based on the received data; identifying, from the high precision detection data, a set of objects that are classifiable by at least one known classifier; generating high recall detection data based on the received data; identifying from the high recall detection data a set of obstacles; and performing an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles, wherein the operation comprises jointly optimizing the high precision detection data of the objects and the high recall detection data of the obstacles, when the autonomous vehicle is performing fallback maneuvers, and wherein the operation comprises determining a cover value between an obstacle in the high recall detection data and an object in the high precision detection data, by dividing area of intersection of the obstacle and the object by area of the obstacle.
2. The method of claim 1, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is greater than a threshold cover value.
3. The method of claim 2, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data.
4. The method of claim 2, wherein if the cover value is between 1.0 and the threshold cover value, the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.
5. The method of claim 2, wherein the threshold cover value is 0.8.
6. The method of claim 2, wherein if a protrusion on the obstacle is the only difference between the obstacle and the object, the operation comprises associating the protrusion with the object and removing the obstacle from the high recall detection data.
7. The method of claim 1, wherein if the cover value is smaller than the threshold cover value, the operation comprises maintaining respective representations of the obstacle and the object.
8. The method of claim 1, wherein if the cover value is smaller than the threshold cover value and if the obstacle is associated with the object, the operation comprises integrating one or more characteristics of the object into characteristics of the obstacle while maintaining respective

representations of the obstacle and the object.

9. A method of controlling an autonomous vehicle, comprising: detecting and tracking objects or obstacles in an environment of the autonomous vehicle according to the method of claim 1; determining a candidate trajectory for the autonomous vehicle based on the high precision detection data and/or the high recall detection data; and controlling the autonomous vehicle according to the candidate trajectory.

10. The method of claim 9, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is greater than a threshold cover value.

11. The method of claim 10, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data.

12. The method of claim 10, wherein if the cover value is between 1.0 and the threshold cover value, the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.

13. The method of claim 10, wherein the threshold cover value is 0.8.

14. The method of claim 10, wherein if a protrusion on the obstacle is the only difference between the obstacle and the object, the operation comprises associating the protrusion with the object and removing the obstacle from the high recall detection data.

15. The method of claim 9, wherein if the cover value is smaller than the threshold cover value, the operation comprises maintaining respective representations of the obstacle and the object.

16. The method of claim 9, wherein if the cover value is smaller than the threshold cover value and if the obstacle is associated with the object, the operation comprises integrating one or more characteristics of the object into characteristics of the obstacle while maintaining respective representations of the obstacle and the object.

17. A system for controlling an autonomous vehicle, comprising: a set of sensors, configured to receive data that represents objects or obstacles in an environment of the autonomous vehicle; and a processor, configured to: generate high precision detection data based on the received data; identify, from the high precision detection data, a set of objects that are classifiable by at least one known classifier; generate high recall detection data based on the received data; identify from the high recall detection data a set of obstacles; and perform an operation on the high precision detection data of the objects and the high recall detection data of the obstacles, based on a status of the autonomous vehicle or based on one or more characteristics of the objects or the obstacles, wherein the operation comprises jointly optimizing the high precision detection data of the objects and the high recall detection data of the obstacles, when the autonomous vehicle is performing fallback maneuvers, and wherein the operation comprises determining a cover value between an obstacle in the high recall detection data and an object in the high precision detection data, by dividing area of intersection of the obstacle and the object by area of the obstacle.

18. The system of claim 17, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is greater than a threshold cover value.

19. The system of claim 18, wherein the operation comprises removing an obstacle from the high recall detection data if the cover value is about 1.0 or the obstacle is identical to an object from the high precision detection data.

20. The system of claim 18, wherein if the cover value is between 1.0 and the threshold cover value, the step of removing is performed after dilating representation boundaries of the object to encompass the obstacle.

21. The system of claim 18, wherein the threshold cover value is 0.8.

22. The system of claim 18, wherein if a protrusion on the obstacle is the only difference between the obstacle and the object, the operation comprises associating the protrusion with the object and removing the obstacle from the high recall detection data.

23. The system of claim 17, wherein if the cover value is smaller than the threshold cover value,

the operation comprises maintaining respective representations of the obstacle and the object.

24. The system of claim 17, wherein if the cover value is smaller than the threshold cover value and if the obstacle is associated with the object, the operation comprises integrating one or more characteristics of the object into characteristics of the obstacle while maintaining respective representations of the obstacle and the object.
